

Tribhuvan University
Institute of Science and Technology
2083
☆

Bachelor Level / Third Year/ Fifth Semester
Bachelors in Information Technology (BIT 301)
(Web Technology II)

Full Marks: 60

Pass Marks: 24

Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Section A

Long Answer Questions.

Attempt any TWO questions.

(2 × 10 = 20)

1. What is a function? Write a PHP program that contains a function that accepts an integer, find its factorial, and returns the value. [2+8]
2. Define a class called BankAccount with properties for account number and balance, and methods for deposit, withdrawal, and balance display. Write a PHP program that creates an object and demonstrates these methods. [10]
3. Differentiate between GET and POST methods. Design a form with input field name, address, phone number, submit button and display button in HTML and write corresponding PHP and MySQL code to store the user's values after click on submit button and display data after click display button. [3+7]

Section B

Short Answer Questions.

Attempt any EIGHT questions.

(8 × 5 = 40)

4. Discuss PHP data types with suitable examples. [5]
5. Write a PHP program to sort an array of integers in ascending order. [5]
6. Write a PHP program that reads a string, and checks whether it is a palindrome. [5]
7. Write a PHP program using regular expressions to validate an email address and a mobile number of 10 digits starting with 98. [5]
8. What are anonymous functions in PHP? Explain their use cases. [5]
9. Explain conversion between arrays and variables with suitable example. [5]
10. What is a cookie? Write a PHP program demonstrating session management using cookie. [1+4]
11. Explain exception handling in PHP with suitable example. [5]
12. Write a PHP program to read a CSV file containing student records (name, roll, marks) and display its contents in a formatted HTML table. [5]

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Bachelor Level / Third Year/ Fifth Semester
Bachelors in Information Technology (BIT 302)
(Software Engineering)

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Section A

Long Answer Questions.

Attempt any TWO questions.

[2 × 10 = 20]

1. Differentiate between plan-driven and agile development. Describe different principles of agile development methods. List its advantages and disadvantages. [4+3+3]
2. Describe three different types of user testing. Explain the acceptance testing process. [5+5]
3. Differentiate between functional and non-functional requirement. Explain any two UML diagrams with example. [3+7]

Section B

Short Answer Questions.

Attempt any EIGHT questions.

[8 × 5 = 40]

4. Explain different key features associated with version control systems. [5]
5. Define product and process standards. List three reasons to justify the importance of these standards. [3+2]
6. Explain different activities that are carried out during software design. [5]
7. What are the different requirement elicitation techniques? Explain. [5]
8. Explain any one interaction model with an example of your own. [5]
9. List the different levels at which software reuse is possible. Explain any two of them. [2+3]
10. Illustrate the fundamental project activities that are common to all organizations. [5]
11. Explain different fundamental architectural views. [5]
12. Write short notes on [2x2.5=5]
 - a) Pipe and filter pattern
 - b) Internet software engineering

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Bachelor Level / Third Year/ Fifth Semester/ science
Bachelors in Information Technology (BIT 303)
(Information Security)

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Section A

Long Answer Questions.

Attempt any TWO questions.

(2 × 10 = 20)

1. Encrypt the plain text "EVERYTHING AROUND US HAS A PATTERN" using Rail fence cipher using 3 as key. What is Galois Field? Using Extended Euclidean algorithm find the multiplicative inverse of 7 in Z_{10} . [5 + 1 + 4]
2. Distinguish between hash value and digital signature. How does SHA – 1 differ with SHA – 2? Describe the working mechanism of MD5. [2 + 2 + 6]
3. Which one is more vulnerable, phishing or backdoors? Justify with some use cases. Describe the countermeasure concepts for malwares. [5 + 5]

Section B

Short Answer Questions.

Attempt any EIGHT questions.

(8 × 5 = 40)

4. What is threat? Describe the types of attacks. [2 + 3]
5. Define two factor authentication. List the security issues for user authentication. [2 + 3]
6. Explain the role based and attribute based access control. [5]
7. List the dimensions for authentication principles. How do you define credential for access right? [3 + 2]
8. Explain the components for security risk analysis. [5]
9. Describe the major cyber offenses under cyber law of Nepal. [5]
10. Define privacy. Describe the different layers of security auditing architecture. [1 + 4]
11. Explain the methods of key generation and encryption / decryption process in RSA. [5]
12. Define attack surface and attack tree. What is trust framework? [3 + 2]

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Bachelor Level / Third Year / Fifth Semester/science
Bachelors in Information Technology (BIT 304)
(Computer Graphics)

Full Marks: 60

Pass Marks: 24

Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Section A

Long Answer Questions.

Attempt any TWO questions.

(2 × 10 = 20)

1. What is projection? Explain parallel and prospective projection with suitable diagram. Illustrate the window to viewport transformation with example. [1+3+6]
2. Write down algorithm steps of mid-point ellipse drawing algorithm. Trace a circle with radius $r = 3$ centred at the origin for the first octant. [4+6]
3. Differentiate between Liang Barsky Line Clipping algorithm and Cohen Sutherland Line Clipping Algorithm. Clip the polygon A (100,150), B (200,250) and C (300,200) with clipping window defined by the coordinates (100,300), (300,300) and (200,100) using Sutherland Hodgeman Polygon Clipping Algorithm. [3+7]

Section B

Short Answer Questions.

Attempt any EIGHT questions.

(8 × 5 = 40)

4. Trace the points on the line path between A (1, 2) and B (7, 8) using Bresenham's line drawing algorithm. [5]
5. Differentiate between boundary and flood fill algorithm with algorithm. [5]
6. What is virtual reality? Explain some form of virtual reality. [2+3]
7. Explain hidden surface with Z – Buffer algorithm. [5]
8. Explain polygon surface representation using Polygon Table and Polygon Meshes. [5]
9. What is raster scan display and random scan display? Explain advantage of each in brief. [2+3]
10. What do you understand by Solid Modeling? Explain Wireframe and Sweep Representation in brief. [1+4]
11. How a Realistic Image can be generated in Computer Graphics? Compare Phong Shading with Gourard Shading. [2+3]
12. Write short notes on: [2x2.5=5]
 - a) Bezier Curve
 - b) OpenGL

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Bachelor Level / Third Year / Fifth Semester
Bachelors in Information Technology (ENG305)
(Technical Writing)

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Section A

Long Answer Questions.

Attempt any TWO questions.

(2 × 10 = 20)

1. Workplace communication requires a strict adherence to ethical standards. Detail the ethical issues surrounding 'Plagiarism' and 'User Safety/Product Liability Warnings' in technical manuals. How can a technical communicator advocate for the user's safety when corporate deadlines or budgets favor cutting corners? Illustrate your arguments with realistic scenarios and potential real-world repercussions.
2. Modern IT enterprises rely heavily on remote, collaborative workflows. Discuss how a technical writer optimizes collaboration within cross-functional teams across the following areas:
 - a) Managing feedback loops and conflict resolution during peer reviews
 - b) Overcoming language barriers and idiomatic expressions in global documentation
 - c) Establishing strict protocols for document branching, merging, and version tagging
3. A leading software company based in Lalitpur has advertised an opening for the post of IT Officer / System Administrator.
 - a) Construct a professional resume highlighting your technical skills, academic projects, and qualifications.
 - b) Write a cover letter to the Hiring Manager explaining why you are the best fit for this position.

Section B

Short Answer Questions.

Attempt any EIGHT questions.

(8 × 5 = 40)

4. You are administration officer in an office. The office has recently announced 'Call for Quotation' for the purchase of stationery items. Submit a recommendation letter to the CEO of your office by comparing and contrasting the quotations.
5. How does the concept of 'audience analysis' differ when writing a technical user manual versus writing a research paper for a scientific journal?
6. How is technical writing different from imaginative (creative) writing and academic writing?

7. Write a persuasive slack message or email to your project manager using the AIDA format, pitching the adoption of an expensive, automated automated-testing tool for your current software project.
8. Define ethnocentrism. What specific steps can a technical communicator take to prevent this bias from isolating international readers in an installation guide?
9. Visual elements should never be mere decorations. Discuss the role of tables and flowcharts in technical documents. What criteria make a flowchart effective?
10. What is a Gantt Chart? Draw a sample Gantt chart representing your academic preparation and revision schedule for the final examinations.
11. Answer the following grammatical and stylistic questions as instructed:
 - a) Identify the error: The committee has disagreed on their individual recommendations for the software patch.
 - b) Choose the correct pronoun: Give the cryptographic key to (whoever/whomever) is authorized to access the server.
 - c) Correct the misplaced modifier: Wrapped in bubble wrap, the technician opened the fragile motherboard.
 - d) Identify the vague pronoun: Rajiv told Dinesh that he was responsible for the database failure.
 - e) Replace the inappropriate pronoun: A professional must always double-check their source code before deployment.
12. Write short notes on:
 - a) Intellectual Property (IP) vs. Open Source Licensing
 - b) Ethical leadership